IQP Generative Models for Strong Gravitational Lensing Images

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Abstract

Instantaneous-quantum-polynomial (IQP) circuits are believed to be hard to sample classically [4, 5], while the expectation values a maximum-mean-discrepancy (MMD) loss needs stay computable, so an IQP Born machine can be trained classically and deployed on quantum hardware [14]. A recent correlation–complexity map, two indices computed from a dataset's Walsh order spectrum and its multi-information, predicts which datasets suit such models, and on it DeepLense strong-lensing images are the most IQP-compatible dataset we measured ($I_{\mathrm{QCLI}} = 0.47$, $I_{\mathrm{CCI}} = 0.87$), above the turbulence data of the paper that introduced it. Given *learned* latent directions, $\theta = \theta^0 + \Phi^\top z$, a 10-qubit IQP Born machine trained on lens crops beats an RBM and a DCGAN on the paper's PDF–JS and on flux-aligned Conv2D-MMD^2 at $N = 11$ and $N = 100$ training images with 2–39$\times$ fewer parameters, a different arm winning each metric, and reconstructs unseen images at correlation 0.97. Three caveats bound it. The reference paper's own fixed latent protocol yields only noise on images, a structure-matched permutation null still scores $I_{\mathrm{QCLI}} = 0.42$, and the DCGAN still leads on raw Conv2D-MMD^2 at $N = 100$. Everything is exact classical simulation at a single seed and $n = 10$ qubits, so we claim no quantum advantage.

1 Introduction

IQP circuits are commuting-gate circuits whose output distribution is the squared Walsh–Hadamard transform of a phase function. Exact classical sampling from them would collapse the polynomial hierarchy [4], and the hardness is conjectured to survive approximate sampling in the average case [5, 6], though sufficient noise restores classical simulability [13]. The same commuting structure keeps an MMD loss's expectation values computable, so such a model can be *trained classically and deployed on quantum hardware* [14, 15].

Whether real data sits where that is useful is an empirical question, and a recent proposal [10] approaches it with two ingredients. (i) A *correlation–complexity map* placing a dataset on two coordinates, the quantum correlation-likeness indicator I_{QCLI}, how much of its Walsh order spectrum sits at high order as an IQP output would, and the classical correlation-complexity indicator I_{CCI}, how much of its total correlation the best pairwise tree misses [18, 16]; the high-I_{QCLI}, high-I_{CCI} corner is argued to be where IQP circuits do well. (ii) A latent-adaptation protocol (Algorithm 1 of the reference paper, which trains a shared phase core on one anchor sample, then adapts a fixed block of 50 Walsh-basis phases taken in Hamming order sample by sample, and interpolates that block to reach unseen samples). Both were demonstrated on one dataset, 2D turbulence snapshots [9] at $n = 18$ qubits.

We take both to simulated dark-matter strong-lensing images from DeepLense [1, 2] and report three results. (1) Lens crops land at the most IQP-compatible position of any dataset we measured. (2) A

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10-qubit IQP Born machine trained on those crops beats the paper's own RBM and DCGAN baselines on PDF–JS and on flux-aligned Conv2D-MMD^2, a different arm winning each metric, with far fewer parameters. (3) It does so only once the fixed Walsh latent block of the reference protocol is made *learnable*. Appendices B–D take the same tests back to the paper's own data, where its turbulence result does not reproduce under our pre-registered convention and two internal claims come out mixed.

2 Setup

Data and encoding. We use the DeepLense `Model II` simulation set (80,194 train / 8,910 validation images, 64×64, three substructure classes pooled, with per-class results in App. F) [1, 3, 11]. Each image is centroid-aligned by an integer shift (offset 3.4 ± 1.6 px) and cropped to the central 32×32 window, keeping 99.4% of the light, and the crop becomes a probability mass function (pmf) $p(k_x, k_y)$ over 1,024 cells on $n = 10$ qubits (App. A). All 2^{10} generators Z_s are used, so $\theta \in \mathbb{R}^{1024}$ and q_θ follows from two length-1024 fast Walsh–Hadamard transforms.

Latent parameterisations. We compare the reference protocol against (i) its fixed 50-phase block trained *jointly* across images and (ii) our *learned-direction* model (a trainable matrix Φ whose M rows are directions in phase space, so that each image becomes a point z_i in M dimensions), written

$$\theta_i \;=\; \theta^0 + \Phi^\top z_i, \qquad \theta^0 \in \mathbb{R}^{1024}, \quad \Phi \in \mathbb{R}^{M \times 1024}, \quad z_i \in \mathbb{R}^M, \tag{1}$$

with $M \in \{2, 4, 8\}$, each row of Φ L^2-normalised to $\sqrt{1024}$, and $\theta^0, \Phi, \{z_i\}$ trained jointly under an L^2 penalty on z of 10^{-3}, raised to 0.3 for the $M{=}8$ arm. The three objectives are the paper kernel (the reference paper's multi-bandwidth Gaussian MMD kernel), the flat MMD (uniform weight on every Walsh coefficient, which makes it estimable from hardware shots) and exact KL (a simulation-only diagnostic, reported in App. E). For unconditional generation z is drawn from a fitted Gaussian or resampled from the trained z_i with jitter.

Baselines and metrics. We compare only against the two baselines the reference paper uses, a restricted Boltzmann machine (RBM; 10 visible and 1,000 hidden units, persistent contrastive divergence with one Gibbs step [8, 17]) and a deep convolutional generative adversarial network (DCGAN) [12], both trained on the same data and both sized in Table 1. Metrics are the paper's two, the probability-density Jensen–Shannon divergence between intensity histograms (PDF–JS) and Conv2D-MMD^2, a biased MMD estimate in the feature space of a fixed random three-layer convolutional encoder [7]. Every IQP arm and the RBM share the flux scalar S while the DCGAN learns its own scale, so raw Conv2D-MMD^2 is partly a brightness comparison and we also report a flux-aligned variant, flagged as post-hoc. Every arm generates 200 images scored against the same 200 held-out images at the reference training-set sizes $N \in \{11, 100\}$, with a single seed (seed 0) throughout.

Departures from the reference setup. Six choices differ from the reference paper, and each follows from the data rather than from the model. Its snapshots carry a time index and its task is to interpolate to unseen times, whereas lens images are static and fall into three substructure classes, so we generate unconditionally from the latent prior and report the classes separately in App. F. It encodes a sample as 18-bit $(x, y, \text{intensity})$ tuples on 18 qubits with the generators truncated by subset size to about 31k phases, whereas we encode the whole crop as one pmf on 10 qubits and keep every generator, because that crop already holds nearly all the light while a raw intensity axis carries barely more than one bit, so a tuple encoding would spend qubits on intensity when the spatial structure is what has to be learned. It fits a core on one anchor and then adapts a fixed block of 50 Walsh phases one sample at a time, whereas we learn the directions Φ and train the N images jointly, since fixed Walsh directions cross-fade pixels instead of moving a ring (Sec. 3.2). It estimates its MMD from 500 model samples per step, whereas at $n = 10$ we evaluate it in closed form. We keep its metrics and its baselines, the RBM reading the same 10-bit code and the DCGAN the same 32×32 crops, add a held-out projection correlation, and paste each generated crop back into 64×64 at a fixed flux $S = 124.92$ and a sampled offset before scoring.

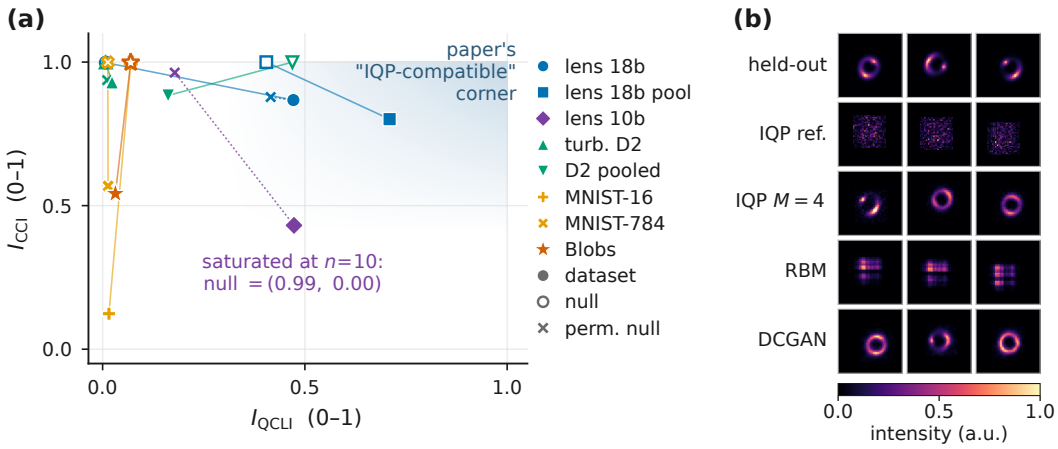


Figure 1: **(a)** Correlation–complexity map, with datasets filled, i.i.d.-uniform nulls at the same (n, K) open and labelled null, and permutation nulls as crosses labelled perm. null (ours, not in the reference paper). The legend also shortens lens 18b for the 18-bit per-image encoding, lens 18b pool for its pooled variant, lens 10b for the 10-bit crop pmf, turb. D2 and D2 pooled for the reference paper's turbulence data, and Blobs for BinaryBlobs. **(b)** First three unconditional samples per model, $N{=}100$, no selection, shared 0–1 colour scale (at most 13 pixels per row are clipped), where IQP ref. is the reference protocol and IQP $M{=}4$ our flat-MMD arm.

3 Results

3.1 Lens images are the most IQP-compatible dataset on the map

Encoded as the reference paper encodes turbulence, at $K = 4{,}096$ sampled tuples per image, the lens images of Fig. 1(a) score $I_{\mathrm{QCLI}} = 0.4715$ and $I_{\mathrm{CCI}} = 0.8677$, higher in I_{QCLI} than the paper's own turbulence data at the same K (0.023), binarised MNIST (0.014) or BinaryBlobs (0.032, both at $K = 10^4$), and inside the map's IQP-compatible corner.

A permutation null that keeps each image's intensity histogram and destroys all spatial structure still scores $I_{\mathrm{QCLI}} = 0.4154$ and *raises* I_{CCI} to 0.8782, leaving only $+0.056$ in I_{QCLI} from lens structure, and the paper's own turbulence data behaves the same way, so at this encoding the map largely tracks the intensity histogram. The 10-bit crop pmf we train on does separate from its permutation null in both coordinates (I_{QCLI} 0.473 against 0.178, I_{CCI} 0.431 against 0.963), and we therefore read a favourable map position as necessary rather than sufficient.

3.2 Why the reference protocol needs learned directions

Run sequentially on our low-variance latent block, Algorithm 1 produces noise (Fig. 1(b), second row; the Hamming-ordered variant is in App. E). Representability is not the cause, since the full 1,024-phase circuit fits a *single* aligned crop to correlation 0.99997 under the flat MMD and 0.9962 under the paper kernel. The protocol is. With the core frozen on one anchor, the 50 fixed latent coordinates reconstruct every other training image at correlation 0.47, because fixed Walsh directions act as an XOR-shift interference pattern that cross-fades pixels instead of moving lens structure, which suits a drifting time series and not an i.i.d. image set.

3.3 Learned latent directions generalise

Equation (1) repairs this. To test generalisation we freeze θ^0 and Φ and optimise only z on 200 *held-out* images. At $N{=}100$ the best arm, flat MMD at $M{=}4$, reaches mean correlation 0.968 (minimum 0.851) against a nearest-training-image reference of 0.970, while the paper kernel reaches only 0.74–0.79. The gain comes from learning the directions and not from the extra parameters, since in the controlled $N{=}20$ sweep of Fig. 2(a) a trained Φ reaches 0.911 against 0.672 for a random Φ of the same size. Useful M stays small, collapsing to 0.095 by $M{=}8$ once the required $\|z\|$ exceeds the model's coherence radius ($\approx$ 0.03–0.05).

Table 1: Paper-style comparison on lens crops, IQP against the two baselines the reference paper uses. Lower is better and **bold** marks the best entry in a column. Learned-direction parameter counts are $N{=}11$ / $N{=}100$, and $\mathrm{MMD}^2_{\mathrm{al}}$ is the flux-aligned variant (post-hoc). The last row is an empirical floor from scoring one held-out half against the other, not a model. Single seed.

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	$\mathrm{MMD}^2_{\mathrm{al}}$	PDF–JS	MMD^2	$\mathrm{MMD}^2_{\mathrm{al}}$
IQP, paper protocol (Alg. 1)	1,024	0.00609	0.2031	0.1837	0.00697	0.2140	0.2233
IQP, fixed basis, joint	1,024	0.00133	0.1416	0.1141	0.00099	0.1041	0.0617
IQP, learned dir. $M{=}4$ (flat MMD)	5,164 / 5,520	0.00196	0.1134	**0.0279**	0.00239	0.0861	**0.0275**
IQP, learned dir. $M{=}4$ (paper kernel)	5,164 / 5,520	**0.00095**	0.1168	0.0679	**0.00092**	0.1189	0.0876
RBM [17]	11,010	0.00632	**0.0741**	0.1240	0.00195	0.0752	0.1144
DCGAN [12]	199,837	0.00183	0.2165	0.1241	0.00763	**0.0371**	0.0694
Held-out split-half (floor)	n/a	*0.00003*	*0.0040*	*0.0069*	*0.00003*	*0.0040*	*0.0069*

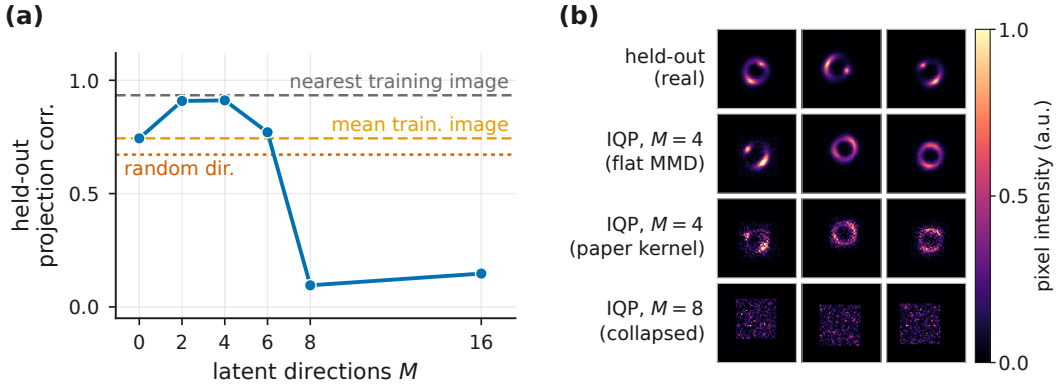


Figure 2: Panels (a) and (b) are different experiments. **(a)** Controlled latent-width sweep ($N{=}20$ mechanism test), held-out projection correlation against the number of learned directions M, with its own reference levels (0.934 nearest training image, 0.744 mean training image, 0.672 random directions); the $M{=}8$ and 16 points are the best over a learning-rate sweep, i.e. favourable to the model. **(b)** Round-3b arms at $N{=}100$, held-out crops on top, then $M{=}4$ under the flat MMD, $M{=}4$ under the paper kernel, and the collapsed $M{=}8$ arm, which was trained with KL and a z-penalty.

3.4 Paper-style comparison and data scaling

In Table 1 the $M{=}4$ paper-kernel arm wins PDF–JS at both training-set sizes (0.00095 and 0.00092) and the $M{=}4$ flat-MMD arm wins the flux-aligned Conv2D-MMD^2 at both (0.0279 and 0.0275), with $2\times$ fewer parameters than the RBM and $39\times$ fewer than the DCGAN, though no single arm wins both metrics. Two of the four IQP arms fall behind the DCGAN on PDF–JS at $N{=}11$ and two fall behind the RBM at $N{=}100$, and on raw Conv2D-MMD^2 the IQP arms *lose* to the RBM at $N{=}11$ (0.0741 against 0.1134) and to the DCGAN at $N{=}100$ (0.0371 against 0.0861), an ordering that reverses once flux is aligned. The paper's data-scaling claim holds only in part, since our DCGAN is *better* at $N{=}11$ than at $N{=}100$ on PDF–JS, the opposite of the low-data GAN collapse the paper reports (Fig. 5, App. E). Trained on each substructure class on its own, the same arm reaches the same level (App. F).

4 Discussion and limitations

The map places lens images where IQP models should do well and the runs bear that out on both metrics, one arm for each, but only with learned latent directions. Nothing here is evidence of quantum advantage, because every circuit is simulated exactly at $n = 10$ qubits and the hardness results describe the model class at scale rather than our instances. All numbers are single seed, and the DCGAN's $3\times$ swing in Fig. 5 sets the scale of the effects we discuss. The imposed flux scalar leaves raw Conv2D-MMD^2 partly a brightness metric and the aligned variant post-hoc, and zero-learning and PCA controls match or exceed the IQP arms on some physical-quantity metrics.

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A Pipeline

Figure 3 summarises the encoding, the latent parameterisation and the evaluation.

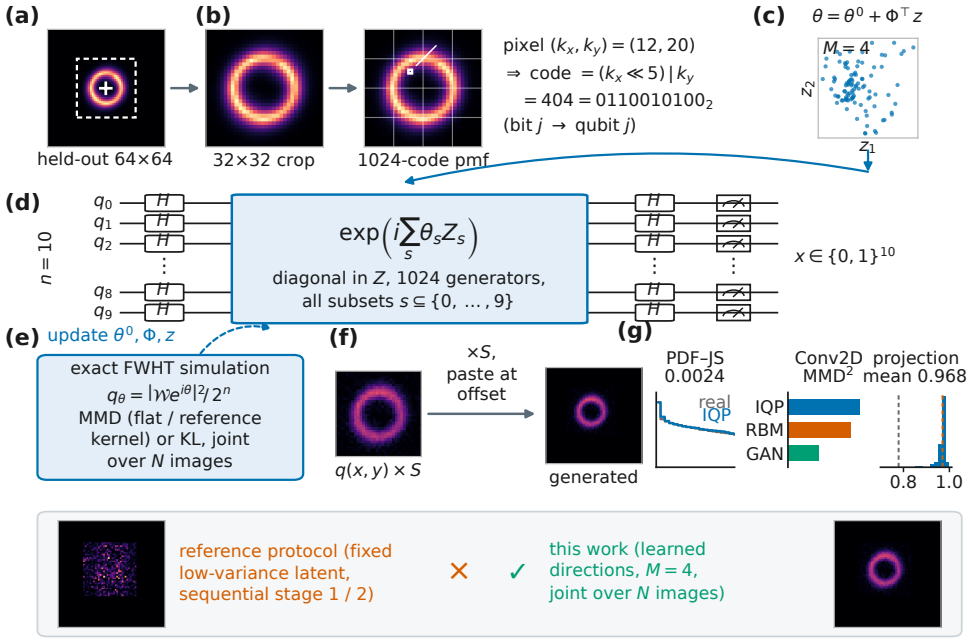


Figure 3: Pipeline. **(a)** A held-out 64×64 image, white cross at its intensity centroid and dashed box the 32×32 crop window. **(b)** The aligned crop read as a pmf over 1,024 cells, where the example pixel $(k_x, k_y) = (12, 20)$ carries code $404 = 0110010100_2$ and bit j addresses qubit j. **(c)** The first two learned latent coordinates of the 100 training images at $M = 4$. **(d)** The IQP circuit, an H layer, one diagonal phase $\exp(i \sum_s \theta_s Z_s)$ over all 1,024 subsets s, a second H layer and measurement. **(e)** Exact simulation through $q_\theta = |\mathcal{W}e^{i\theta}|^2/2^n$, trained jointly over the N images by MMD (flat or reference kernel) or by KL, the dashed arrow updating θ^0, Φ and z. **(f)** A trained $q(x, y)$ scaled by S and pasted at a sampled offset. **(g)** The intensity PDF of real against IQP samples (PDF–JS 0.0024), Conv2D-MMD^2 at N=100 (IQP 0.086, RBM 0.075, DCGAN 0.037), and projection correlations over 200 held-out images (mean 0.968). The bottom row contrasts a reference-protocol sample with an M=4 sample from this work.

B Replication of the reference paper's turbulence experiment

We attempted a direct replication on the reference paper's own dataset (2D cylinder-wake turbulence D2 [9], $n = 18$ qubits, 11 anchor snapshots). Under our pre-registered evaluation convention the result is **not reproduced**, since in the 11-snapshot regime the RBM and a 1.2M-parameter DCGAN beat our IQP on *both* of the paper's metrics. Two findings qualify this. Under a last-write-wins aggregation convention, never specified in the paper, our IQP's PDF–JS values do land close to the published ones (0.0322 and 0.0476 against 0.0238 and 0.0221), but the RBM still wins. Conv2D-MMD^2 is also saturated there, since ground truth shuffled in pixel space already scores 0.704, below every model in the paper's Table 2 (0.90–1.47), and the unseen-time synthesis task is solvable by linear interpolation (correlation 0.9998). Our anchor-fit ceiling (correlation 0.69 sampled, 0.77 exact) traces to a frozen θ_{lat} in stage 1 (≈ -0.08), a 5×10^4-sample pmf (≈ -0.04) and a 0.88–0.89 model-family ceiling, not to our implementation.

C Support mismatch (reference paper Fig. 3)

The reference paper's Fig. 3 claims (a) that the upper envelope of achievable MMD falls as the target's I_{QCLI} rises, because the support-induced floor shrinks, and (b) that the envelope drops overall as

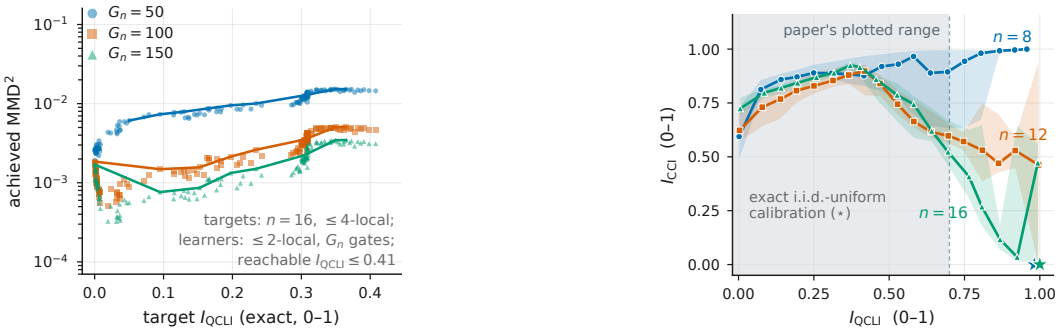


Figure 4: **(a)** Support-mismatch study at $n = 16$, with 140 target circuits ($\leq$ 4-local) spanning exact $I_{\mathrm{QCLI}} \in [0, 0.408]$ fitted by $\leq$ 2-local learners of 50, 100 and 150 gates (exact MMD, Adam, 5,000 steps). More gates lower the whole envelope (claim (b), reproduced), while within the reachable range the achieved MMD^2 *rises* with target I_{QCLI}, the opposite of claim (a). **(b)** Circuits optimised to maximise I_{QCLI} only, $n = 8, 12, 16$, 3 seeds and 2 gain settings per n, 3,333 steps each and about 20,000 logged points per n. I_{CCI} never enters the objective.

the learner gets more gates (Fig. 4(a)). Claim (b) reproduces, as the median achieved MMD^2 falls $9.57 \times 10^{-3} \to 2.15 \times 10^{-3} \to 1.45 \times 10^{-3}$ across the three learners. Claim (a) runs the other way inside the range we can reach, since the achieved MMD^2 *rises* as the target's exact I_{QCLI} goes from 0 to 0.41 (Spearman $\rho = +0.98/+0.90/+0.81$), a trend that also holds for a control trained on exact pmfs. The paper's premise still holds mechanistically ($\rho \leq -0.98$ for the Proposition II.2 floor proxy), but here the achieved MMD is dominated by trainability rather than by support mismatch. Our target family saturates at exact $I_{\mathrm{QCLI}} \approx 0.41$, and at 0–0.064 when estimated from 10^4 samples as the paper does, so the regime where claim (a) is asserted lies outside what we could construct.

D The I_{QCLI}–I_{CCI} ceiling

We also reproduced the experiment in which maximising I_{QCLI} alone raises I_{CCI} as a by-product (Fig. 4(b)). The coupling reproduces at low I_{QCLI} (Spearman $\rho = +0.83/+0.76/+0.95$ for $n = 8/12/16$ restricted to $I_{\mathrm{QCLI}} \leq 0.3$) but reverses above $I_{\mathrm{QCLI}} \approx 0.38$ for $n = 12, 16$ (global $\rho = -0.53, -0.57$). The reversal is forced by the definitions, because the exactly uniform distribution maximises I_{QCLI} (0.982/0.998/1.000 for $n = 8/12/16$) and has $I_{\mathrm{CCI}} = 0$. The reference paper's figure stops below $I_{\mathrm{QCLI}} \approx 0.7$, so a high I_{QCLI} need not indicate exploitable structure and can simply indicate proximity to uniform.

E Additional arms

Data scaling. At $N \in \{10, 30, 50, 70, 90, 100\}$ the learned-direction IQP arm is flat for $N \geq 30$ (raw Conv2D-MMD^2 0.082–0.087) and so is the RBM (0.066–0.089), while the DCGAN leads already at N=10 (0.044 against 0.120) and swings by 3$\times$ non-monotonically (0.113 $\to$ 0.037), as Fig. 5 shows.

Exact KL. KL needs the full simulated distribution and cannot be estimated from hardware shots, so we keep it out of Table 1 and record it here. Its M=2 arm, on 3,094 / 3,272 parameters, reaches PDF–JS 0.00175 at N=11 and 0.00302 at N=100, with raw Conv2D-MMD^2 0.1673 and 0.1356 and flux-aligned 0.0559 and

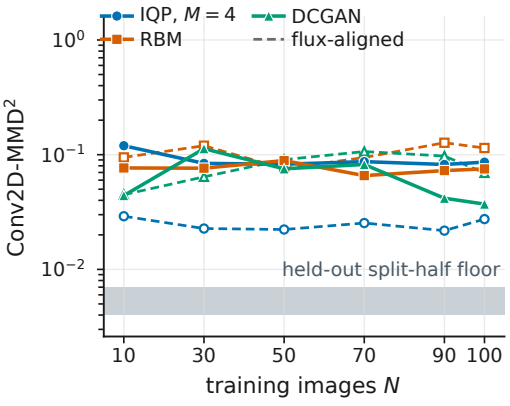


Figure 5: Conv2D-MMD^2 against training-set size, raw as solid lines and flux-aligned as dashed lines with open markers, over the band of the held-out split-half floor.

0.0866. The collapsed $M=8$ arm of Fig. 2(b) is also a KL arm.

Prior draws at $M=2$. All draws from the $M=2$ prior are rings of rms radius 9.40 ± 0.21 px against 9.21 ± 0.15 for real data, though under-dispersed in ellipticity (0.048 against 0.099).

Other arms. We ran 25 conditions, 16 IQP arms, the RBM and DCGAN baselines and seven zero-learning or PCA controls. The arms outside Table 1 (Hamming-ordered latent block, other objectives and priors, a pseudo-time conditional arm) and the controls are scored under the same metrics in the released tables. Trained jointly rather than sequentially, the Hamming-ordered latent block of the reference protocol reaches PDF–JS 0.00137 at $N=11$ and 0.00139 at $N=100$, so what fails in Sec. 3.2 is the sequential protocol and not the choice of block, and we did not run the Hamming block under the sequential protocol. Every IQP arm keeps a nearest-neighbour distance to the held-out set (3.6–5.9) far above the split-half floor (0.95), so none is memorising the training set, and the $M=8$ arm collapses (PDF–JS 0.0103, MMD^2 0.244 at $N=100$).

F Per-class generation

Every result in the main text pools the three substructure classes. To check that pooling is not hiding a class that fails, we retrained the arm of Table 1 ($M=4$, flat MMD, resample prior, seed 0) separately on each class and scored each run against 200 held-out images of that same class, changing nothing else. At $N=100$ all three classes land at the pooled arm's level on every metric of Table 2, and at $N=11$ all three still train, with held-out projection correlations near 0.9. Each class is scored against its own held-out set, which carries its own split-half floor, so the rows compare with their pooled counterpart rather than with each other. The samples in Fig. 6 look alike across classes, which is what the data implies since the three differ in only a few per cent of the pixel variance, and we claim that each class can be generated on its own, not that the model separates them.

Table 2: Per-class generation with the arm of Table 1, each run trained on one class and scored against 200 held-out images of that class, with the pooled arm at the matching N for reference and lower better in every column except the projection correlation.

Training set	N	PDF–JS	MMD^2	MMD$^2_{\mathrm{a1}}$	proj. corr.
axion only	11	0.002197	0.1015	0.02636	0.9232
cdm only	11	0.002174	0.1108	0.02031	0.9238
no sub only	11	0.001102	0.08096	0.02379	0.8998
pooled	11	*0.001963*	*0.1134*	*0.02786*	*0.9092*
axion only	100	0.002050	0.09012	0.02346	0.9720
cdm only	100	0.002275	0.09783	0.01848	0.9686
no sub only	100	0.002001	0.1034	0.02773	0.9703
pooled	100	*0.002387*	*0.08607*	*0.02745*	*0.9684*

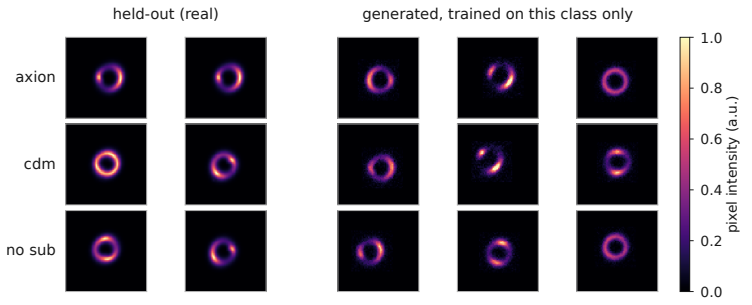


Figure 6: Per-class samples. Each row is one class, two held-out images followed by three generated by the arm trained on 100 images of that class alone, taken as the first two and the first three in index order with no selection, on the shared 0–1 colour scale.